

National College of Ireland

Project Submission Sheet

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Machine Learning for Sustainable Transportation in Ireland: EV Charging Infrastructure and Vehicle Fuel Type Prediction

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1. Abstract — This project is analysis on the shift towards the sustainable transportation in Ireland with both the electric vehicle (EV) infrastructure and fuel usage patterns. In this project there are two Irish datasets: one is an EV charging infrastructure dataset of Ireland, and another one is Irish vehicle registration dataset. The main aim of this project is to analyze EV charging infrastructure and predict the fuel types using multiple machine learning techniques.

This project includes data preprocessing steps such as handling missing values, encoding categorical features, and creating new features to improve data quality. For the vehicle dataset, Extra Trees and CatBoost models were implemented to classify fuel types and for the EV infrastructure dataset the Logistic Regression and Random Forest models were used to predict the operational status of the EV charging stations.

The results clearly show that CatBoost model achieved 91.00% accuracy for the fuel prediction, and the Random Forest model achieved 97.65% accuracy for the EV infrastructure dataset. The project highlights that how machine learning can really support better planning of EV infrastructure and helps to understand the changes in vehicle fuel trends in Ireland over a period.

Keywords — CatBoost, Extra Trees, Logistic Regression, Random Forest, Classification, Feature Engineering, Model Evaluation.

2. Introduction

The transition towards the sustainable transportation has become one of the main focuses across all the countries in this modern world. Where electric vehicles (EV) playing a key role in reducing carbon emissions and dependence on fossil fuels. In Ireland, the adoption of electric vehicles (EV) are growing constantly and it is mainly supported by the government policies, environmental awareness, and advancements in charging technology. But still the effectiveness of this transition is not only depend on the electric vehicles (EV) it also depends on the availability and reliability of the charging infrastructure.

The main challenge in this domain is the unreal gap between infrastructure development and actual vehicle demand. While the charging stations are being installed across Ireland, their distribution, operational status, and

accessibility not always perfectly align with the charging patterns of vehicle fuel usage. In the meantime, traditional fuel types such as petrol and diesel still dominate in vehicle registrations which clearly indicates that the transition to clear energy is gradual rather than immediate implementation.

This project is motivated by the need for data-driven insights, which gives far more understanding of both infrastructure readiness and vehicle fuel trends. Machine learning techniques provide effective approach to analyze complex datasets and identify clear patterns that may not be easily visible through traditional analysis.

The main objectives of this project are fourfold:

- To analyze EV charging infrastructure in Ireland and to understand its distribution.
- To predict the operational status of charging stations using machine learning techniques.
- To predict vehicle fuel types based on the registration data.
- To compare the multiple machine learning models performance and with that we can evaluate the effectiveness of model for transportation related data analysis.

3. Related Work

In most recent years, the machine learning has been widely used in the field of sustainable transportations, especially for analysing the electric vehicles (EVs), which is helpful in predicting energy consumption, and understanding the vehicle trends. Many of the researchers have focused on the improvement for data-driven decision making. For example, Ahsan et al. [4] explored how the machine learning can be applied to the EV energy management and showed that the predictive models are helpful in improving efficiency and optimise energy usage. Similarly, Hussain et al. [5] compared the different machine learning algorithms which is used for predicting energy consumption in EVs and found that the ensemble models tend to perform the task better because they can handle complex data patterns more effectively and efficiently.

Apart from the infrastructure and the energy-related studies, the machine learning has also been used for vehicle classification tasks. Divya et al. [1] used machine learning

techniques to classify the vehicle fuel types based on sales data and showed that supervised learning models can successfully capture the fuel usage patterns. This is relevant to the current study, as it also focuses on predicting fuel types using structured vehicle data. In addition, Seto et al. [8] demonstrated that the advanced models such as gradient boosting can easily outperform simpler models like Logistic Regression, especially when working with larger datasets.

Another important aspect of the highlighted part of the literature is the need for proper evaluation and handling of imbalanced datasets. Techniques such as cost-sensitive learning [2] and the use of advanced evaluation metrics like F1-score and ROC-AUC [3] have been recommended by most of the researchers to improve the model performance and to ensure the more reliable result. The linear models are often used as baseline approaches because of their simplicity, although it may not always provide the best performance compared to more complex methods [7].

Moreover, the ensemble-based models such as Extra Trees and CatBoost models have gained the popularity in recent years due to their strong performance in classification tasks [9], [10]. These models are particularly useful when dealing with the datasets that contain a mix of numerical and categorical features, as they can capture non-linear relationships more effectively and efficiently.

However, most of the existing research focuses on either EV-related analysis or vehicle classification individually. There is very limited work that combines both the infrastructure analysis and the fuel type prediction in a single study. This project mainly aims to address that gap by integrating both the perspectives by providing a more data-driven understanding of the sustainable transportation trends in Ireland.

4. Methodology

4.1 Data Collection

This project is done by using two real world datasets which is related to vehicle in Ireland. Here the two datasets are different but related in perspectives.

The first dataset focuses completely focuses on the vehicle registration patterns and the fuel type, And the second dataset completely focuses on the EV charging infrastructure.

The first dataset consists of the Irish vehicle registration data, which includes the information such as fuel type, vehicle category, registration period, and the total registration values. This dataset is used to analyze the fuel usage trends and to develop models for predicting the vehicle fuel types.

The second dataset which is EV charging infrastructure dataset, it contains the more detailed information of the charging stations including the operational status, operator details, number of charging points, geographical coordinates, connection types, charging power, and the date related fields clearly. This EV dataset is primarily used to understand the readiness and the distribution of the EV infrastructure, as well as to helps in building the models for predicting the operational status of the charging stations.

4.2 Data Preprocessing

The data preprocessing was performed separately for both the dataset, because they are differing significantly in the structure, format, and modelling requirements.

For the vehicle registration dataset, the data was first examined for the missing values, duplicate records, and inconsistencies. But the dataset was almost clean, with no major missing value issues. However, the necessary validation steps were performed out to ensure the data quality. Categorical variables such as fuel type, vehicle category, and registration type were properly encoded for model compatibility.

In addition to that, the feature engineering was performed to improve the model performance. Time-based features such as year, month, and quarter were extracted from the registration period to capture the temporal patterns. Numerical transformations such as the logarithmic and square root scaling of registration values were applied to handle the skewness in the data distribution of the dataset. The dataset was also goes through filter process to focus on the major fuel classes (such as petrol and diesel), which simplified the classification task and improved model stability far better.

For the EV charging infrastructure dataset, the original JSON data was first flattened into the structured tabular format to make it perfect fit for analysis. The relevant features such as charging station status, operator information, region, number of charging points, latitude, longitude, connection types, and the power-related attributes were extracted properly.

The missing numerical values were handled carefully by using appropriate imputation methods (such as median imputation), while the categorical values were filled using the most frequent category. Furthermore, several irrelevant and leakage prone features including IDs, URLs, direct status indicators, and the metadata fields were removed before the model training. This step was very crucial, because we need to ensure that the models do not learn from the unintended or redundant information.

At the end of preprocessing, we can ensure that both datasets were clean, consistent, and suitable for the machine learning, but also it helps in improving the ability of the models to capture the meaningful patterns.

4.3 Exploratory Data Analysis

The Exploratory Data Analysis (EDA) was performed mainly to understand the structure, distribution, and key patterns in both datasets before building the models.

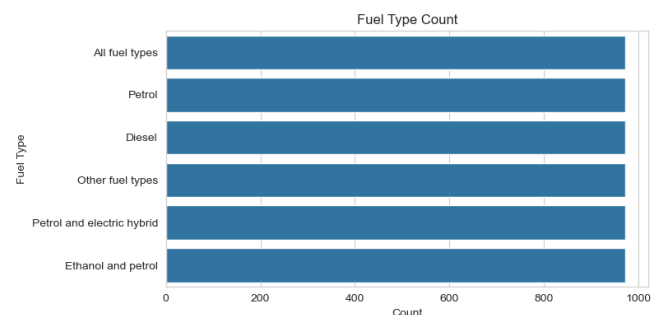


Figure A.1: Distribution of Vehicle Fuel Types

This figure A.1 shows the clear proportion of different fuel types in the dataset. It is clearly observed that the petrol and the diesel dominate the registrations, while alternative fuel types have a relatively smaller share overall.

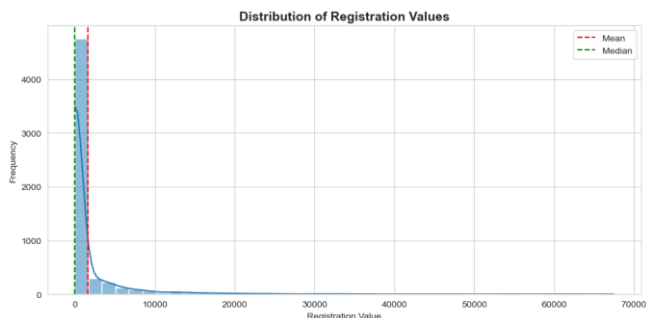


Figure A.2: Distribution of Vehicle Registration Values

The figure A.2 plot clearly shows, how the distribution of registration counts across the different categories. The data appears to be highly skewed, with very few categories having significantly higher values.

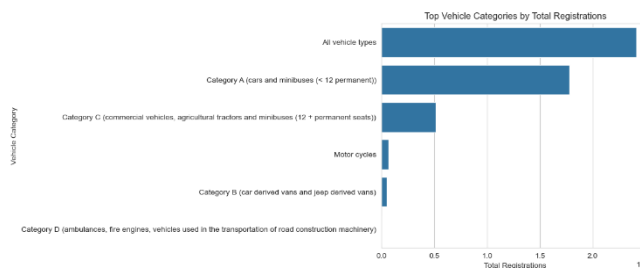


Figure A.3: Top Vehicle Categories by Total Registrations

The figure A.3 chart highlights the most common vehicle categories based on the total registrations recorded in the dataset. It helps to identify which types of vehicles have most demand in the Irish market.

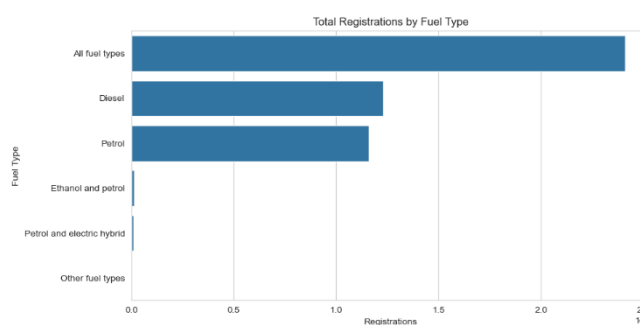


Figure A.4: Total Vehicle Registrations by Fuel Type

The figure A.4 clearly compares the total number of registrations across the different types of fuel, which again clearly shows the dominance of traditional fuel vehicles.

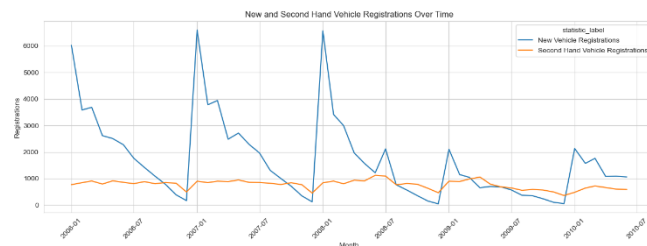


Figure A.5: New vs Second-Hand Vehicle Registrations Over Time

The figure A.5 plot clearly shows the trend between new and second hand vehicle registrations in Ireland. Which provides the clear insight about the Irish consumer purchasing behaviour over the time.



Figure A.6: Yearly Vehicle Registrations by Fuel Type

The figure A.6 clearly shows how fuel preferences changes over a period. It helps us in understanding the gradual transitions in fuel usage.

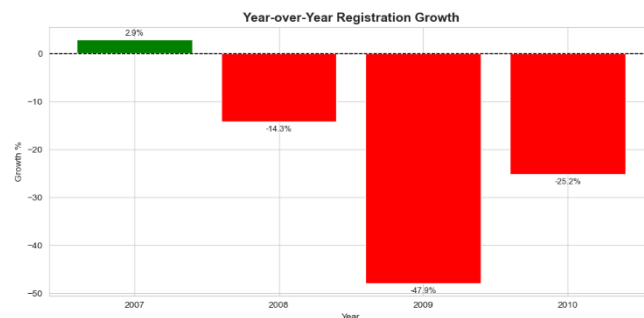


Figure A.7: Year-over-Year Registration Growth

The figure A.7 chart clearly represents the growth rate of vehicle registrations across the years, highlighting the fluctuations and trends.

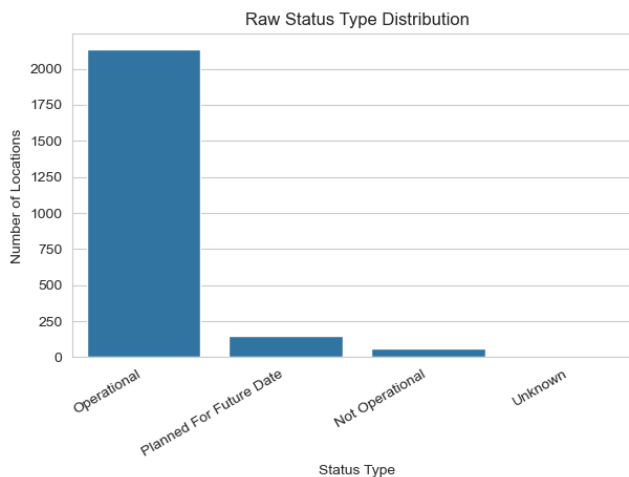


Figure B.1: Distribution of Charging Station Status Types

The figure B.1 clearly shows the distribution of different status categories of EV charging stations. Most stations are operational, indicating strong infrastructure availability.

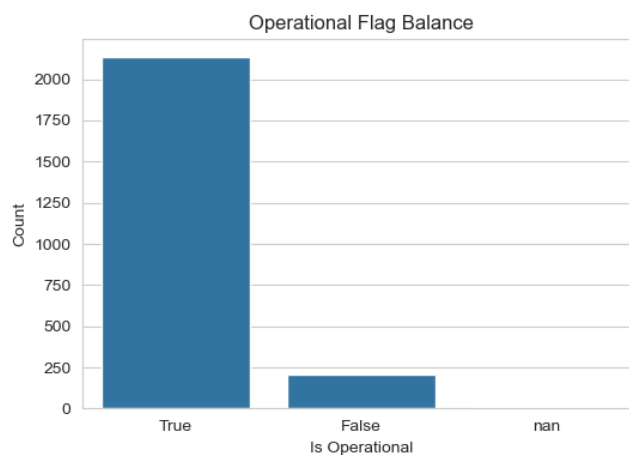


Figure B.2: Balance of Operational vs Non-Operational Charging Stations

The figure B.2 is simplified plot which highlights the class imbalance in the dataset, where the operational chargers dominate.

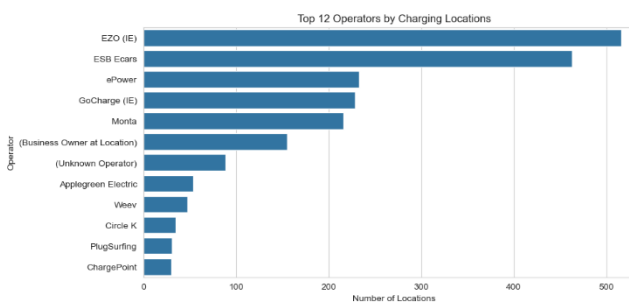


Figure B.3: Top EV Charging Operators by Number of Locations

The figure B.3 chart clearly identifies the major charging operators in Ireland, which shows that few operators control a large portion of the infrastructure in the Irish market.

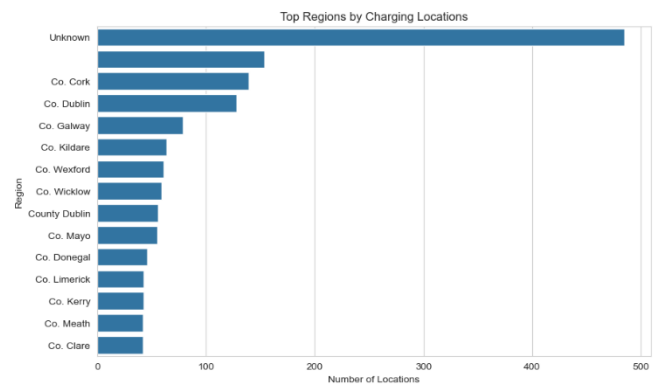


Figure B.4: Regional Distribution of EV Charging Stations

The figure B.4 clearly shows that how the EV charging stations are distributed across different regions in the Ireland, which helps to identify areas with higher or lower infrastructure availability as of now.

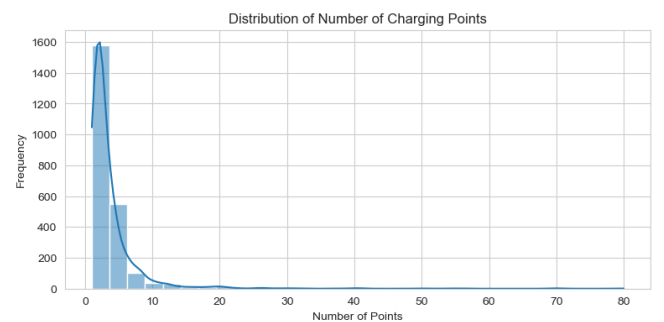


Figure B.5: Distribution of Charging Points per Location

The figure B.5 plot clearly shows the number of charging points available at each station, indicating that most locations have limited capacity in Ireland.

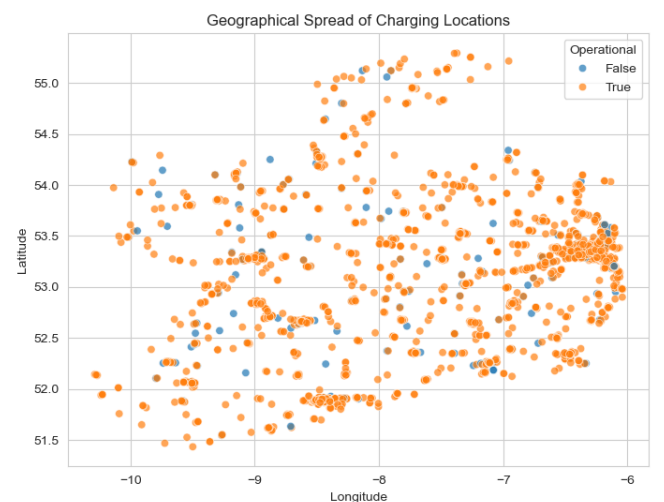


Figure B.6: Geographical Distribution of Charging Stations

The figure B.6 map visualisation highlights the spatial spread of charging stations, showing that the urban regions holds higher concentration.

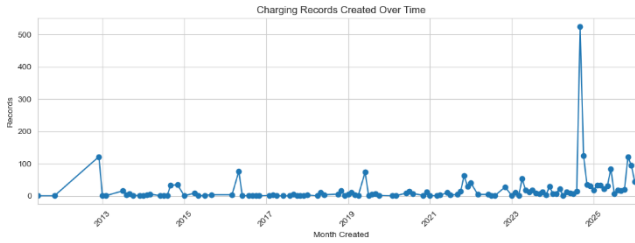


Figure B.7: Charging Station Records Over Time

The figure B.7 clearly shows how charging station records have increased over time, indicating growth in the EV infrastructure.

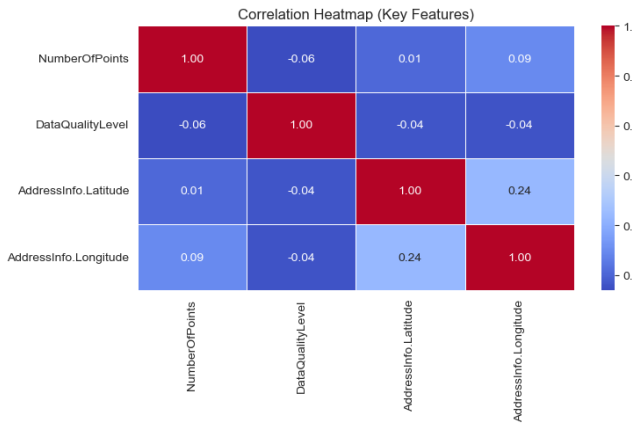


Figure B.8: Correlation Heatmap of EV Infrastructure Features

The figure B.8 heatmap clearly shows the relationships between key numerical features, indicating weak linear correlations and supporting the use of ensemble models.

4.4 Model Building

The supervised machine learning models were developed for both the datasets to perform classification tasks. For the vehicle registration dataset, the CatBoost and the Extra Trees Classifier were implemented to predict vehicle fuel types. CatBoost model was selected because of its strong performance with the categorical features and its ability to manage and handle them efficiently without extensive preprocessing requirements. Extra Trees Classifier model was used due to its robustness and the capability to handle the high dimensional structured data, which makes it suitable for capturing the complex patterns in the vehicle registration data.

And for the EV charging infrastructure dataset, the Logistic Regression and the Random Forest models were implemented to classify the operational status of the EV charging stations. The Logistic Regression model was used as a baseline model due to its simplicity and interpretability, because it provides the basic understanding of relationships between features. The Random Forest, on the other hand, was

used as an ensemble model, because it is capable of capturing the complex, non-linear relationships and interactions between the multiple features.

4.5 Model Improvement and Hyperparameter Tuning

The vehicle dataset is gone through the feature engineering to improve the model learning. The time based features such as year, month, and quarter were created to capture the trends in vehicle registrations of Ireland. The registration value feature was highly skewed, so that the transformations which is log and square root scaling were applied properly to make the data more balanced structure.

The hyperparameter tuning was performed for both the CatBoost and the Extra Trees models. The parameters such as number of trees, depth, and the learning rate were adjusted to improve the accuracy and to reduce the overfitting of the models. To ensure the reliable model performance, The stratified cross validation was used, which helped to maintain the class balance and provided the better generalisation.

The EV infrastructure dataset also gone through feature engineering because it mainly focused on the connection and power related attributes. The new features such as connection count, maximum power, mean power, DC charging indicator, and the fast charging indicator were created to represent charging station characteristics even better. The categorical features were encoded properly, and the leakage prone features such as IDs and the direct status related fields were removed before training the model.

The hyperparameter tuning was mainly applied to the Random Forest model, where the parameters such as number of trees and the depth were adjusted to improve the performance. The logistic Regression model was used as a baseline model. Since the dataset was imbalanced in nature the class weighting was considered to reduce bias towards the majority class, and it improves the prediction of minority classes.

5. Results and Discussion

This section represents the performance of the machine learning models for both the datasets, along with a detailed comparison and interpretation of the results. The evaluation is based on the standard classification metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.

5.1 Vehicle Fuel Type Dataset

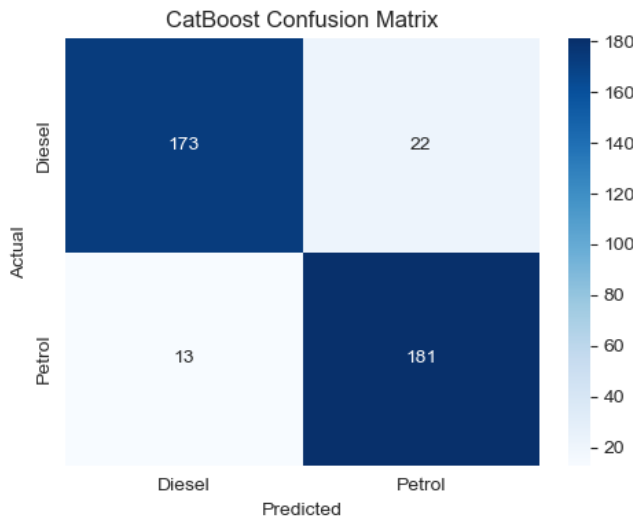


Figure A.8: CatBoost model confusion matrix

The figure A.8 clearly shows that the model correctly classified 173 diesel instances and 181 petrol instances, while the misclassifying 22 diesel as petrol and 13 petrol as diesel. This indicates a high level of the classification accuracy with relatively low error rates across both the classes.



Figure A.9: Extra Trees model confusion matrix

This figure A.9 clearly shows that the Extra Trees model have 166 diesel instances, and 186 petrol instances were correctly classified. But still the model misclassified 29 diesel as petrol and 8 petrol as diesel, which is slightly higher compared to the CatBoost model in terms of total errors.

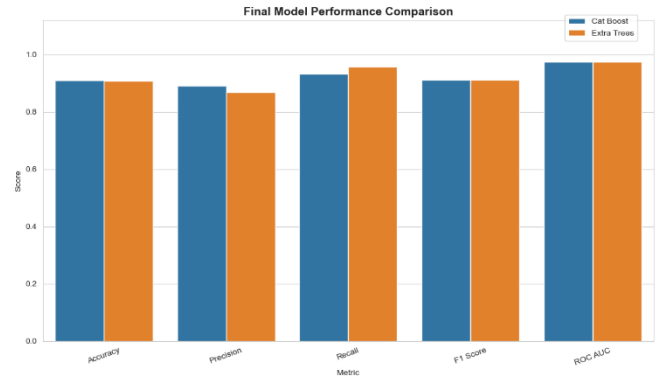


Figure A.10: Final model performance comparison

The overall performance comparison of both the model is presented in Figure A.10. The Model performance comparison was between the CatBoost vs Extra Trees model. The CatBoost model achieved an accuracy of **91.0%**, while The Extra Trees model achieved an accuracy of **90.5%**. In terms of the precision, the CatBoost model recorded approximately **0.912**, compared to **0.909** for the Extra Trees model. The recall values were **0.911** for the CatBoost model and **0.909** for the Extra Trees model. Similarly, the F1-score for the CatBoost model was **0.911**, while the Extra Trees model was achieved **0.909**. The ROC-AUC scores were also high for both the models, with CatBoost model at approximately **0.975** and the Extra Trees model at **0.974**.

In terms of model stability, the CatBoost model achieved a cross validation standard deviation (CV Std) of **0.0133**, while the Extra Trees model achieved **0.0138**. The slightly lower standard deviation of the CatBoost model clearly indicates more consistent performance across different folds compared to the Extra Trees model. Overall, both the models demonstrate a strong and consistent performance across all the evaluation metrics, with the *CatBoost model* showing slightly better results in terms of accuracy and balanced classification.

5.2 EV Infrastructure Dataset

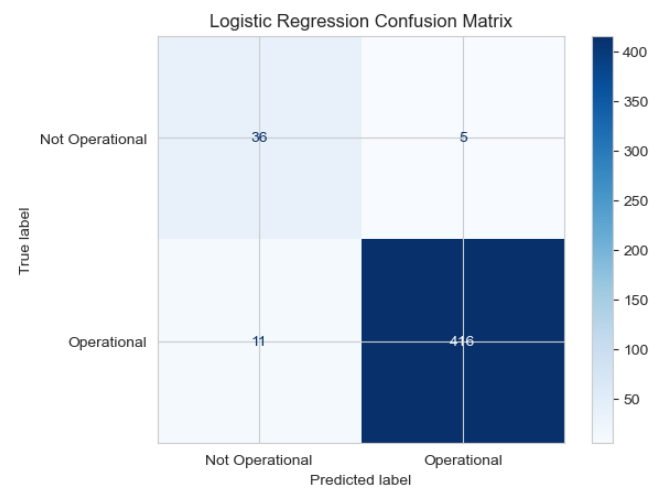


Figure B.10: Logistic Regression model confusion matrix

The figure B.10 clearly shows that the Logistic Regression model has a slightly higher number of misclassifications, particularly in identifying the minority

class. Although the model performs reasonably well, it is less effective compared to the Random Forest model.

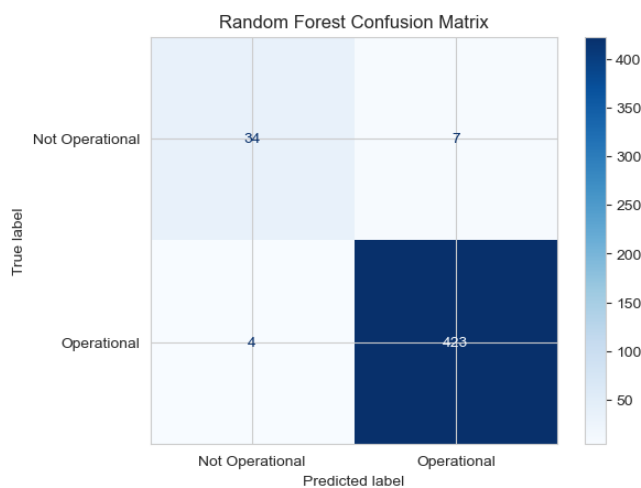


Figure B.11: Random Forest model confusion matrix

The figure B.11 clearly shows that the Random Forest model, this model correctly classified majority of the charging stations across both classes and it has very few misclassifications. The number of correctly predicted operational and non-operational chargers is significantly high which indicates that this model has strong model performance.

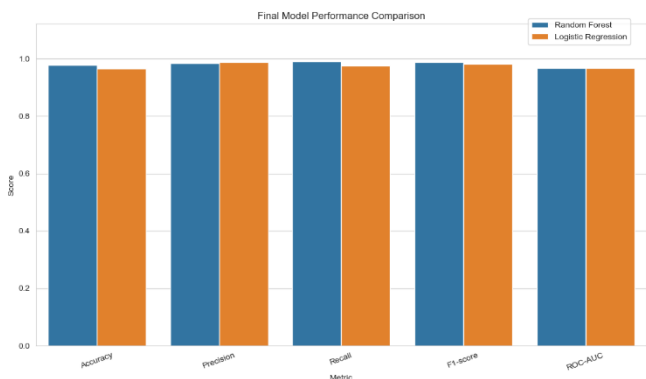


Figure B.12: Final model Performance Comparison

The figure B.12 clearly shows that the model performance comparison of the Logistic Regression model and the Random Forest model. The Random Forest model achieved an accuracy of **97.65%**, while the Logistic Regression model achieved **96.58%**. The precision values were approximately **0.987** for the Random Forest model and **0.981** for the Logistic Regression model. The recall values were **0.987** and **0.981**, respectively. The F1-scores were **0.987** for the Random Forest model and **0.981** for the Logistic Regression model. The ROC-AUC scores were also high, with the Random Forest model at approximately **0.9678** and the Logistic Regression model approximately **0.9674**.

In terms of the model stability, the Random Forest model achieved a cross validation standard deviation (CV Std) of **0.009016**, while the Logistic Regression model achieved **0.009005**. The standard deviation values for both the models are low it clearly indicates that the consistent performance across different folds, with the *Random Forest model* maintaining slightly better overall in terms of performance.

6. Conclusion and Future Work

6.1 Conclusion

The analysis clearly shows several important insights into sustainable transportation in Ireland. First, the vehicle fuel type prediction shows that strong and consistent patterns exist in registration data, which makes it suitable for the reliable classification using the machine learning models. However, the dominance of petrol and the diesel indicates that the transition towards cleaner fuel types is still gradual in growth. From the infrastructure perspective, the EV charging stations are largely operational and well established, this suggests that the foundational support for the EV adoption is already in place. Despite this, the continued reliance on traditional fuels highlights a clear gap between the infrastructure availability and the actual usage behaviour.

Another main observation was that the ensemble learning models consistently performed better across both the datasets, indicating that the presence of non-linear relationships in transportation data. This clearly suggests that the real world transport systems are more complex and require more advanced modelling approaches to get accurate analysis. Overall, this study highlights the importance of analysing both demand (vehicle behaviour) and the supply (infrastructure) together, rather than in isolation. It shows that the current EV infrastructure was insufficient to Ireland.

6.2 Future Work

The future work can be more focus on incorporating the additional features such as socio-economic and the regional data to improve the model accuracy. The use of explainable AI methods has ability to provide better interpretability of model decisions. Further, applying time series forecasting and the geospatial analysis can help in predicting the future trends and helps to identify the areas where infrastructure development is required more.

7. References

[1] Divya, M & Akshith, Bommoju & Keshava, B & Pranay, Midivelli & Sai, Gangavarapu & Kumar, Bodasu & Shruthi, Shruthi. (2025). Machine Learning for Fuel Type Classification: Insights from the 2023 Telangana sales dataset. International Journal of Engineering Research and Science & Technology.21.30013008.10.62643/ijerst.2025.v21.n2.pp3001-3008. J. Wu, “Statistical Approach for AML Threshold Tuning with Python,” Medium, Aug. 17, 2022. https://www.researchgate.net/publication/400120163_MACHINE_LEARNING_FOR_FUEL_TYPE_CLASSIFICATION_INSIGHTS_FROM_THE_2023_TELANGANA_VEHICLE_SALES_DATA_SET

[2] “Cost-Sensitive Learning for Imbalanced Classification,” Machine Learning Mastery, Jan. 14, 2020. <https://machinelearningmastery.com/cost-sensitive-learning-for-imbalanced-classification/>

- [3] R. Tiwari, "Advanced Evaluation Metrics for Imbalanced Classification Models," Medium (CueNex), Feb. 9, 2023.
<https://medium.com/cuenex/advanced-evaluation-metrics-for-imbalanced-classification-models-ee6f248c90ca>
- [4] Ahsan, Md Shameem & Rahman, Md Khaledur & Islam, Md & Talukder, Md Jakaria & Islam, Md Nazmul. (2025). Machine Learning Methodologies for Electric Vehicle Energy Management Strategies. SSRN Electronic Journal. 4. 99-118. 10.2139/ssrn.5271481.
https://www.researchgate.net/publication/391807671_Machine_Learning_Methodologies_for_Electric_Vehicle_Energy_Management_Strategies
- [5] Hussain, Izhar & Ching, Kok & Uttraphan, Chessda & Gaik, Tay & Noor, Adil. (2025). Evaluating machine learning algorithms for energy consumption prediction in electric vehicles: A comparative study. Scientific Reports. 15. 10.1038/s41598-025-94946-7.
https://www.researchgate.net/publication/391571548_Evaluating_machine_learning_algorithms_for_energy_consumption_prediction_in_electric_vehicles_A_comparative_study
- [6] Baziar, M., Yousefi, M., Oskoei, V. *et al.* Machine learning-based prediction of heating values in municipal solid waste. *Sci Rep* **15**, 14589 (2025).
<https://doi.org/10.1038/s41598-025-99432-8>
- [7] Lin, Yu-Chen & Chen, Si-An & Liu, Jie-Jyun & Lin, Chih-Jen. (2023). Linear Classifier: An Often-Forgotten Baseline for Text Classification. 1876-1888. 10.18653/v1/2023.acl-short.160.
https://www.researchgate.net/publication/372919038_Linear_Classifier_An_Often-Forgotten_Baseline_for_Text_Classification
- [8] Seto, H., Oyama, A., Kitora, S. *et al.* Gradient boosting decision tree becomes more reliable than logistic regression in predicting probability for diabetes with big data. *Sci Rep* **12**, 15889 (2022).
<https://doi.org/10.1038/s41598-022-20149-z>
- [9] GeeksforGeeks, "Machine Learning - Extra Trees Classifier for Feature Selection", 2025.
<https://www.geeksforgeeks.org/machine-learning/ml-extra-tree-classifier-for-feature-selection/>
- [10] GeeksforGeeks, "Machine Learning - Cat Boost", 2025.
<https://www.geeksforgeeks.org/machine-learning/catboost-ml/>